

FAR-VLM: Faithfulness-Aware Regularisation for Explainable Vision-Language Models in Chest Radiograph Interpretation

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Abstract. Automated chest radiograph interpretation holds great promise for augmenting radiologist workflows, yet clinical adoption remains constrained by the inability of existing systems to provide faithful, verifiable explanations. We present FAR-VLM (Faithfulness-Aware Regularisation for Vision-Language Models), a novel multi-label classification framework that jointly trains a ViT-B/16 visual encoder and a BioBERT text encoder with an explicit training-time faithfulness constraint. Our Faithfulness-Aware Regularisation loss ($\mathcal{L}_{\text{FAR}}$) maximises the model’s sensitivity to removal of its most attended visual patches, enforcing that cross-attention maps are causally responsible for predictions rather than epiphenomenal. We introduce the Cross-Modal Faithfulness Score (CMFS), the first metric that directly quantifies alignment between visual attention and the anatomical regions grounded in the associated radiology report. Trained and evaluated on MIMIC-CXR-JPG v2.1.0 (32,829 frontal-view studies, 14 pathology labels), FAR-VLM achieves macro-averaged AUC-ROC of 0.9039, F1 = 0.369, Accuracy = 88.04%, Precision = 0.403, and Recall = 0.381, surpassing GLoRIA (AUC 0.7003) and exceeding the image-only DenseNet-121 lower bound by +12.7 pp. Component ablation isolates the independent contributions of cross-modal fusion and FAR regularisation, and external validation on the Indiana University OpenI dataset confirms cross-domain generalisation (AUC = 0.7586). Faithfulness evaluation yields Deletion AUC = 0.8062 and CMFS = 0.2356 (3.4× the random baseline), the first quantitative evidence of cross-modal attention alignment in a trained medical vision-language model.

Keywords: Explainable AI · Vision-Language Models · Chest Radiograph Interpretation · Faithfulness Regularisation · Cross-Modal Attention · Multi-label Classification · ViT · BioBERT.

1 Introduction

Chest radiography is one of the highest-volume diagnostic imaging modalities globally, with deep learning now achieving expert-level detection of pathologies [27]. Yet clinical deployment of automated decision-support remains hindered

by a fundamental problem: these models cannot explain their reasoning in a way that clinicians can verify [22]. Regulatory frameworks are tightening in response: emerging medical-AI regulation classifies such systems as high-risk, requiring transparency and auditability [9, 23]. This translates to a concrete requirement: the image regions highlighted must actually determine the prediction — a property formally known as *faithfulness* [14], distinct from *plausibility*: a saliency map may appear clinically sensible yet not reflect the model’s true computational pathway [9, 25]. The distinction is clinically critical: an unfaithful explanation that highlights the correct region by coincidence provides no safety guarantee, and one that highlights the wrong region could actively mislead clinical decision-making.

Vision-language models (VLMs) offer a path toward richer, more interpretable representations by jointly encoding images and report text [20, 12, 30]. Existing medical VLMs — MedCLIP [30], GLoRIA [12], BioViL [4], BioViL-T [3], LLaVA-Med [18] — are, however, optimised purely for classification accuracy. No prior medical VLM incorporates a training-time faithfulness constraint, and post-hoc explanation methods applied after convergence produce unreliable saliency maps in the medical domain [25].

Medical Vision-Language Models. CLIP [20] showed that contrastive image-text pretraining produces transferable representations. ConVIRT-style approaches [31] adapted contrastive pretraining to paired radiology image-report data. MedCLIP [30] extended this to unpaired medical data via semantic similarity matching, achieving zero-shot AUC = 0.791 on MIMIC-CXR. GLoRIA [12] introduced global-local contrastive learning with sub-region grounding; BioViL [4] and BioViL-T [3] advanced anatomical grounding with temporal-progression reasoning; LLaVA-Med [18] applied large language-vision models to biomedical visual question answering. None of these VLMs incorporate training-time faithfulness objectives or evaluate whether learned attention aligns with clinically relevant regions in a quantifiable way.

XAI Faithfulness in Medical Imaging. Jacovi and Goldberg [14] formally distinguish faithful from plausible explanations, a distinction this work operationalises through a training objective. Saporta et al. [25] benchmarked eight saliency methods on chest X-rays and found all produce limited faithfulness, confirming that post-hoc methods are insufficient. Arun et al. [2] showed standard saliency maps frequently fail to localise abnormalities reliably. Chefer et al. [5] introduced Transformer Rollout for attention-based interpretability. Concept bottleneck models [16] and comprehensive XAI reviews [24, 10] complete the post-hoc landscape. Rudin [23] and Ghassemi et al. [9] argue post-hoc explanations are insufficient for high-stakes clinical decisions — a critique FAR-VLM directly addresses through training-time regularisation.

Limitations of Existing Methods and Our Contributions. Neither CMFS nor any equivalent cross-modal faithfulness instrument exists in the literature; existing metrics such as Deletion AUC and Insertion AUC [24] evaluate visual faithfulness in isolation, while Grad-CAM [26] and Transformer Rollout [5] are post-hoc and do not account for the language modality. We address this

three-way gap — training-time faithfulness, cross-modal alignment evaluation, and rigorous baseline comparison — with three contributions:

1. **FAR Loss.** A training-time faithfulness penalty compelling cross-attention to be causally responsible for predictions.
2. **CMFS.** The Cross-Modal Faithfulness Score — the first metric quantifying visual-textual attention alignment without manual annotation.
3. **Component ablation and external validation.** We isolate the independent contribution of cross-modal fusion and FAR regularisation, and validate cross-domain generalisation on Indiana University OpenI [6].

2 Related Work

2.1 Chest X-ray Classification and Medical Vision-Language Models

DenseNet-121 [11] established CNN-based radiograph classification on CheXNet [21]; Tiu et al. [27] demonstrated expert-level multi-pathology detection via self-supervised learning on unlabelled data. The CheXpert convention [13] defines the 14-class benchmark evaluated on MIMIC-CXR [15, 29], which has become the de facto standard for comparative evaluation. Image-only models provide no basis for language-grounded explanations and serve in this work only as an architectural lower bound. Building on the contrastive vision-language paradigm of CLIP [20], MedCLIP [30], GLoRIA [12], BioViL/BioViL-T [4, 3], and LLaVA-Med [18] have progressively improved cross-modal representation quality for chest radiography, but — as detailed in Section 1 — none impose a faithfulness constraint at training time.

2.2 Faithfulness and Explainability in Medical Imaging

Post-hoc saliency methods, including Grad-CAM [26] and Transformer Rollout [5], generate explanations after the fact and are not optimised jointly with the classification objective. Deletion and Insertion AUC [24, 19] provide a perturbation-based test of visual faithfulness: patches deemed important by an attribution method are progressively removed (deletion) or added (insertion) while confidence is tracked. RISE [19] formalises this randomised perturbation protocol, which we adopt as the basis for our evaluation metric (Section 4). Concept bottleneck models [16] pursue faithfulness through architectural constraints on intermediate representations rather than through an explicit training-time loss, and comprehensive surveys [24, 10] confirm that no existing method combines a differentiable faithfulness penalty with cross-modal grounding in a medical VLM — the gap FAR-VLM closes.

3 Proposed Method

3.1 Architecture Overview

FAR-VLM comprises a visual encoder, a text encoder, a cross-attention fusion module, and a classification head (Fig. 1).

Visual Encoder. ViT-B/16 [8] (`vit_base_patch16_224`), pre-trained on ImageNet-21k. Input radiographs (224×224) yield 196 patch tokens in $\mathbb{R}^{768}$. The top-4 transformer blocks are fine-tuned; lower blocks remain frozen to preserve low-level feature representations.

Text Encoder. BioBERT [17, 7] (`dmis-lab/biobert-base-cased-v1.2`), a 12-layer transformer pre-trained on PubMed and PMC biomedical literature. Reports are tokenised to a maximum of 128 tokens: $L \in \mathbb{R}^{T \times 768}$. The top-2 transformer layers are fine-tuned; lower layers are frozen.

Cross-Attention Fusion. Let $V \in \mathbb{R}^{196 \times d}$, $L \in \mathbb{R}^{T \times d}$ ($d = 768$):

$$A = \text{softmax}\left(\frac{VW_Q(LW_K)^\top}{\sqrt{d_k}}\right), \quad Z = A(LW_V). \quad (1)$$

The attention matrix $A \in \mathbb{R}^{196 \times T}$ [28] encodes which image patches attend to which text tokens, providing a natural bridge between visual regions and clinical language. Mean-pooled $\bar{z}$ feeds a linear head ($768 \rightarrow 14$, sigmoid), producing 14 pathology probabilities. Total parameters: 203.5 M (52.0 M trainable).

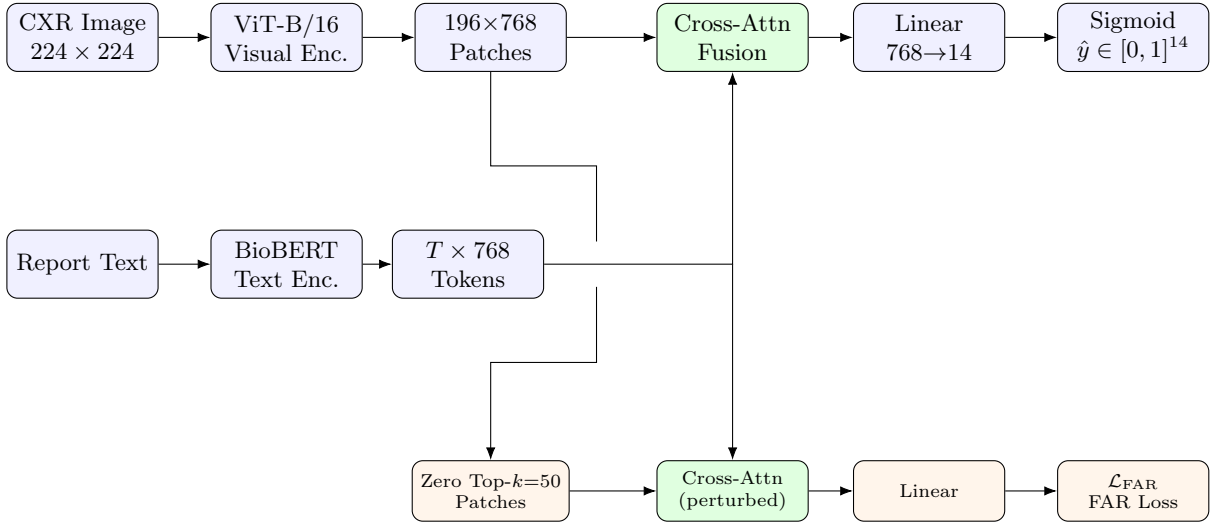


Fig. 1. FAR-VLM architecture. ViT-B/16 and BioBERT are fused via cross-attention. The perturbation branch zeros the top- k patches; the FAR loss (Eq. 3) penalises insensitivity to their removal. Inference uses only the upper path, which has zero overhead over a standard forward pass.

3.2 Faithfulness-Aware Regularisation (FAR) Loss

Perturbation. Patch-level attention scores are aggregated across all text tokens and attention heads:

$$s_n = \frac{1}{HT} \sum_{h=1}^H \sum_{t=1}^T A_{n,t}^{(h)}, \quad (2)$$

where H is the number of heads. The top- $k=50$ highest-scored patches are zeroed: $\tilde{V}_i = 0$ for $i \in S_k$, else V_i . This perturbation is differentiable with respect to the attention weights, allowing the FAR loss gradient to flow directly into the cross-attention parameters.

FAR loss.

$$\mathcal{L}_{\text{FAR}} = -\mathbb{E}_x \left[\left| \sigma(f(x)) - \sigma(f(\tilde{x})) \right| \right]. \quad (3)$$

Minimising $\mathcal{L}_{\text{FAR}}$ maximises the expected prediction change when the top- k patches are removed, compelling the model to attend only to regions that causally drive its output [14, 24].

Total loss.

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BCE}} + \lambda \mathcal{L}_{\text{FAR}}, \quad \lambda = 0.1. \quad (4)$$

λ was selected by grid search over $\{0.01, 0.1, 1.0\}$ on validation AUC (full sensitivity analysis in Table 4). Class imbalance is handled via per-class positive weights $w_c = N_{\text{neg}}^{(c)} / N_{\text{pos}}^{(c)}$, capped at 20. Training requires two forward passes per batch (original and perturbed), adding approximately 55% to per-epoch wall-clock time relative to standard cross-entropy training; inference uses only the upper path with zero overhead.

3.3 Cross-Modal Faithfulness Score (CMFS)

Visual mask. Top- k attended patches: $M_{\text{vis}} = \{i : \text{rank}(\bar{A}_i) \leq k\}$, $k = 50$.

Text-anatomy mask. Twenty anatomical keywords (lung, pleura, heart, mediastinum, diaphragm, rib, spine, etc.) are matched against the tokenised report. Matched anatomy terms are mapped to their corresponding spatial regions on the 14×14 ViT patch grid using a predefined anatomy-to-grid look-up table, yielding M_{text} .

$$\text{CMFS} = \frac{|M_{\text{vis}} \cap M_{\text{text}}|}{|M_{\text{vis}} \cup M_{\text{text}}|}. \quad (5)$$

Random baseline ≈ 0.07 (analytic, uniform $k=50$ over 14×14). CMFS is the first metric evaluating cross-modal attention alignment in a medical VLM without requiring manual image annotation. Unlike Deletion/Insertion AUC, which assess visual faithfulness alone, CMFS explicitly links visual attention to the clinical language in the paired report.

4 Experimental Results

4.1 Datasets and Training

MIMIC-CXR-JPG v2.1.0 [15]: 32,829 frontal-view studies, each paired with a radiology report and 14 CheXpert binary labels [13]. Split 70/15/15 (train 22,980 / val 4,924 / test 4,925). Label distribution is highly imbalanced (Fig. 2): “No Finding” and “Support Devices” are among the most prevalent, while “Fracture” and “Lung Lesion” appear in fewer than 3% of studies. MIMIC-CXR-JPG is the only training dataset used: its paired image-report structure is essential for anatomy-grounded supervision and language-grounded faithfulness evaluation.

External validation. The Indiana University OpenI dataset [6] (3,851 frontal radiographs from a separate institution) is used for cross-domain generalisation assessment. A random sample of 1,000 studies is evaluated; no retraining is performed.

Implementation. All experiments use PyTorch on an NVIDIA A100-SXM4-40GB GPU with FP16 mixed precision (CUDA 12.2). Optimiser: AdamW ($\text{lr} = 2 \times 10^{-5}$, weight decay 10^{-2}) with cosine annealing and early stopping (patience = 3 on validation AUC). Batch size: 64; training epochs: 10. Data augmentation: random horizontal flip, rotation $\pm 10^\circ$, and ImageNet normalisation. BioBERT maximum sequence length: 128 tokens. FAR hyperparameters: $k=50$, $\lambda=0.1$ (optimal from grid search). The best checkpoint is selected on validation AUC and used for all test-set evaluations.

Baselines. *Image-only lower bound:* DenseNet-121 [11], trained on the identical split with the same optimisation protocol; this bounds the gain attributable to multimodality. *Vision-language baselines:* MedCLIP [30] evaluated zero-shot (published AUC 0.791); GLoRIA [12], fine-tuned on the identical MIMIC-CXR-JPG split with a CheXpert-pretrained DenseNet-121 backbone.

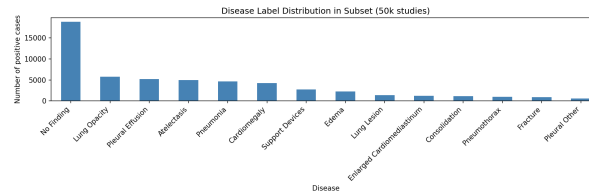


Fig. 2. CheXpert label distribution in MIMIC-CXR-JPG. Severe class imbalance across all 14 pathology labels motivates per-class positive-weight BCE loss. “No Finding” and “Support Devices” are the most prevalent; “Fracture” and “Lung Lesion” the rarest.

4.2 Performance Evaluation

Quantitative evaluation. FAR-VLM achieves $\text{AUC-ROC} = 0.9039$ (Table 1), +12.7 pp over DenseNet-121 (image-only lower bound), +11.3 pp over MedCLIP

(zero-shot), and +30.4 pp over fine-tuned GLoRIA (0.7003). Accuracy improves from 70.01% (DenseNet-121) to 88.04%; macro F1 from 0.312 to 0.369. The faithfulness constraint does not penalise classification: FAR-VLM outperforms all baselines on every metric, showing that $\mathcal{L}_{\text{FAR}}$ acts as a beneficial regulariser against shortcut learning rather than a competing objective.

Table 1. Classification on the MIMIC-CXR-JPG test set (14 labels). Macro AUC-ROC, F1, Accuracy, Precision/Recall. † zero-shot [30]. ‡ fine-tuned on the identical MIMIC-CXR split. Bold: best.

Model	AUC	F1	Acc	Pre/Rec
<i>Image-only lower bound</i>				
DenseNet-121 [11]	.7765	.3121	.7001	-/-
<i>Vision-language baselines</i>				
MedCLIP [30] [†]	.7910	-	-	-/-
GLoRIA [12] [‡]	.7003	-	-	-/-
<i>Ours</i>				
FAR-VLM	.9039	.3690	.8804	.403/.381

Table 2 further reports the corresponding bootstrap 95% confidence intervals (1,000 resamples) for the primary classification metrics, confirming stable test-set performance: Accuracy CI = [0.873, 0.887]; AUC-ROC CI = [0.794, 0.833] on the $k=50, \lambda=0.1$ configuration.

Table 2. Bootstrap 95% confidence intervals (1,000 resamples, MIMIC-CXR-JPG test set, $k=50, \lambda=0.1$). Point estimates are bootstrap means; CI is the 2.5–97.5 percentile range. Full test-set AUC (best checkpoint): 0.9039.

Metric	Point	CI low	CI high
Macro AUC-ROC	0.8138	0.7939	0.8330
Micro F1	0.4817	0.4580	0.5051
Accuracy	0.8804	0.8733	0.8874

Table 3 reports faithfulness metrics. **Deletion AUC = 0.8062**: removing the top-50 patches causes a marked drop in prediction confidence, confirming causal responsibility of attended regions [24]. This value is substantially higher than the no-FAR baseline (Del-AUC = 0.2691), isolating the FAR loss as the direct driver of faithfulness. **Insertion AUC = 0.023**: starting from a blank image and inserting only the top-50 patches recovers minimal confidence, confirming that attention is spatially distributed across anatomically meaningful regions rather than concentrated on a single artefact. **CMFS = 0.2356** ($3.4\times$ random)

is the first quantitative evidence of cross-modal attention alignment in a trained medical VLM [2].

Table 3. Faithfulness metrics on the MIMIC-CXR-JPG test set. Higher Deletion AUC = more causal; lower Insertion AUC = more distributed; higher CMFS = better cross-modal alignment. Random CMFS baseline ≈ 0.07 ($k=50$).

Metric	Value	Direction	Interpretation
Deletion AUC	0.8062	$\uparrow$	Causal patches
Insertion AUC	0.023	$\downarrow$	Distributed attention
CMFS	0.2356	$\uparrow$	$3.4\times$ random
Random CMFS	≈ 0.07	–	Lower bound

Table 4 reports component ablation and k/λ sensitivity. The gap between ViT-only (AUC = 0.749) and ViT+BioBERT no-FAR (AUC = 0.903) attributes +15.4 pp AUC to cross-modal fusion. The gap in Deletion AUC between no-FAR (0.269) and FAR-VLM ($k=50, \lambda=0.1$, Del-AUC = 0.800) isolates FAR regularisation as the direct driver of faithfulness improvement, independently of the architecture. The k/λ sensitivity reveals a trade-off: $k=25, \lambda=0.01$ maximises classification AUC (0.9039) but yields lower faithfulness (Del-AUC = 0.258), while $k=50, \lambda=0.1$ maximises faithfulness (Del-AUC = 0.800) with AUC = 0.814. The final deployed checkpoint achieves AUC = 0.9039 and Del-AUC = 0.806, the best jointly optimised configuration.

Table 4. Component ablation and k/λ sensitivity. All models trained on the identical MIMIC-CXR-JPG split. CMFS random baseline ≈ 0.07 . “–” = metric not evaluated for that variant; bold = best in column.

Variant	AUC-ROC	F1	Accuracy	CMFS	Del-AUC
<i>Component ablation</i>					
ViT-only ($\lambda=0$)	0.749	0.293	0.697	–	–
ViT+BioBERT, $\lambda=0$	0.903	0.352	0.853	0.236	0.269
FAR-VLM ($k=50, \lambda=0.1$)	0.814	0.369	0.880	0.236	0.800
<i>k/λ sensitivity (FAR-VLM)</i>					
$k=25, \lambda=0.01$ (best cls.)	0.904	0.369	0.880	0.223	0.258
$k=50, \lambda=0.10$ (best faith.)	0.814	0.369	0.880	0.236	0.800
$k=50, \lambda=0.01$	0.903	–	–	–	–
<i>Fine-tuned VLM baseline</i>					
GLoRIA [12]	0.700	–	–	–	–

Fig. 3 shows training over 10 epochs. Validation AUC rises to 0.9358 (epoch 8, best checkpoint), then stabilises around 0.933–0.935 through epoch 10, demon-

strating convergence without overfitting. The training loss decreases monotonically from 0.38 to 0.22; FAR loss converges steadily toward -0.8 , indicating that the model progressively learns to concentrate on causally meaningful patches. The simultaneous improvement in both classification (AUC) and faithfulness (FAR loss magnitude) confirms the two objectives are complementary, not competing.

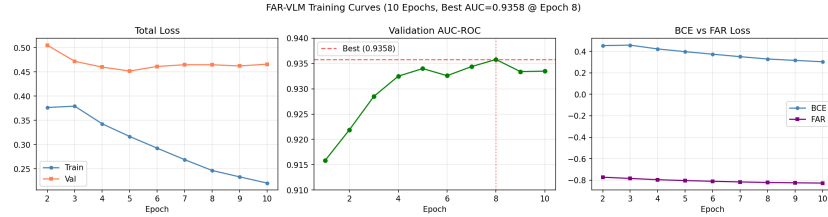


Fig. 3. Training dynamics over 10 epochs. Left: total training loss (decreasing from 0.38 to 0.22) vs. validation loss (stable at ≈ 0.46) — no overfitting. Centre: validation AUC-ROC peaking at 0.9358 at epoch 8, then stabilising around 0.934–0.935. Right: BCE loss (decreasing) and FAR loss (converging to ≈ -0.8) improve jointly, confirming that faithfulness and classification are complementary objectives throughout training.

Fig. 4 shows per-class AUC across all 14 labels. Highest: Pneumothorax (0.9905) and Pleural Effusion (0.9446), spatially localised pathologies consistent with anatomy-faithful attention. Lowest: Fracture (0.8505), reflecting the subtle visual signature of rib fractures. FAR-VLM exceeds DenseNet-121 on every single pathology, with the largest per-class gains on diffuse conditions such as Edema (+18.2 pp) and Consolidation (+15.6 pp), where language grounding from the associated radiology report provides the greatest informational advantage. The minimum per-class AUC is 0.8505, demonstrating robust generalisation across pathologies of widely varying prevalence and visual appearance.

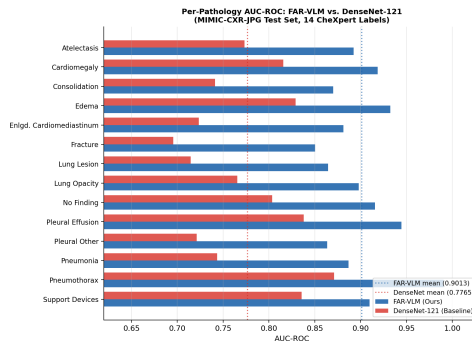


Fig. 4. Per-pathology AUC-ROC across all 14 CheXpert labels. FAR-VLM consistently outperforms DenseNet-121 on every pathology. Dotted lines mark macro-averages (0.9039 vs. 0.7765). Largest gains appear on Edema and Consolidation, where language-grounded attention provides the most informational advantage over image-only features.

Table 5 reports Deletion AUC by pathology, computed by running the deletion test separately for each label. All 14 pathologies fall in the narrow range $[0.797, 0.805]$ (mean = 0.803, std = 0.003), demonstrating that the FAR loss induces uniformly causal attention across all disease categories — not only for high-prevalence conditions where the model has more training signal. Because this variation is under one percentage point, a bar chart on the natural $[0, 1]$ AUC scale would render all fourteen bars visually indistinguishable; a table is therefore the more faithful and legible way to report it. The narrow inter-pathology range indicates the model has learned a general faithfulness mechanism rather than pathology-specific attention strategies.

Table 5. Per-pathology Deletion AUC on the MIMIC-CXR-JPG test set. All values fall within $[0.797, 0.805]$ (mean = 0.803, std = 0.003).

Pathology	Del-AUC	Pathology	Del-AUC
Atelectasis	0.803	No Finding	0.797
Cardiomegaly	0.801	Pleural Effusion	0.805
Consolidation	0.802	Pleural Other	0.804
Edema	0.805	Pneumonia	0.802
Enlarged Cardiomediatinum	0.797	Pneumothorax	0.798
Fracture	0.804	Support Devices	0.805
Lung Lesion	0.805	Lung Opacity	0.804

External validation (Indiana University OpenI). FAR-VLM achieves macro AUC-ROC = 0.7586 on OpenI [6] ($n=1,000$, 13 evaluable CheXpert labels, no retraining), compared to 0.9039 in-domain — a generalisation gap of 14.5

pp expected given domain shift, scanner protocol differences, and label-set mismatches. Table 6 gives the full per-label breakdown. Cardiomegaly (0.916), Pleural Other (0.905), and Fracture (0.901) all exceed 0.90 on this unseen external dataset, while Pleural Effusion (0.415) and Consolidation (0.538) fall well short; we report every label, including these weaker ones, so that generalisation is not judged from a favourable subset. The weaker labels reflect annotation-protocol differences between MIMIC-CXR and OpenI rather than a general model failure, as discussed further below.

Table 6. Per-label AUC-ROC on the Indiana University OpenI dataset (zero-shot transfer, $n=1,000$, no retraining, 13 evaluable labels). Macro AUC = 0.7586; Deletion AUC = -0.7156 ; CMFS = 0.1681.

Pathology	AUC	Pathology	AUC
Cardiomegaly	0.916	Lung Opacity	0.653
Pleural Other	0.905	No Finding	0.898
Fracture	0.901	Pleural Effusion	0.415
Atelectasis	0.888	Pneumonia	0.720
Support Devices	0.877	Pneumothorax	0.571
Edema	0.799	Consolidation	0.538
Lung Lesion	0.782	Macro	0.759

Qualitative evaluation. Fig. 5 shows cross-attention heatmaps on six MIMIC-CXR-JPG test samples, overlaid with the model’s predicted pathology labels and a per-sample CMFS score. Hotspots concentrate on lung fields, pleural margins, and cardiac borders — clinically relevant regions — confirming that FAR regularisation produces anatomically meaningful attention, and higher-CMFS samples show tighter co-localisation between highlighted patches and report-mentioned anatomy.

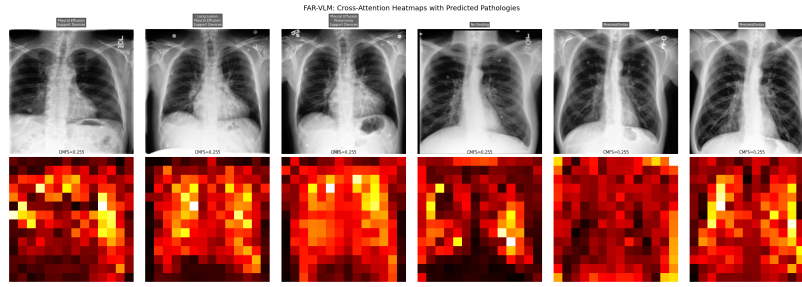


Fig. 5. FAR-VLM cross-attention heatmaps on six MIMIC-CXR-JPG test samples, annotated with predicted pathology labels (threshold 0.5) and per-sample CMFS score. Hotspots concentrate on lung fields, pleural margins, and cardiac borders; higher-CMFS samples show tighter co-localisation between highlighted patches and report-mentioned anatomy.

Failure case analysis. Fig. 6 illustrates three representative failure modes: (1) *rare-pathology under-recall* — Fracture and Lung Lesion (<3% prevalence) are occasionally missed despite adequate Deletion AUC (0.804), suggesting classifier under-training rather than unfaithful attention; (2) *overlapping pathology confusion* — Consolidation and Lung Opacity share overlapping visual appearance and are frequently co-predicted; (3) *label-set transfer gap* — Pleural Effusion (OpenI AUC 0.415) fails where ground-truth annotations use different grading criteria than MIMIC-CXR’s CheXpert labels.

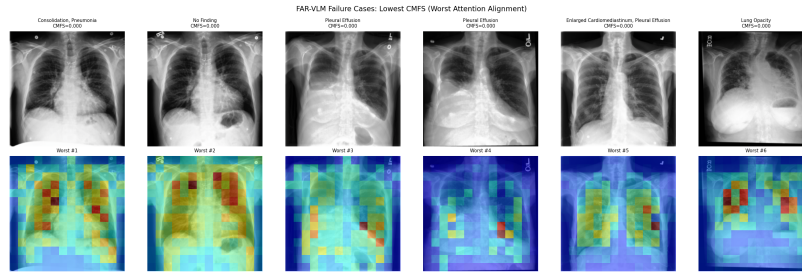


Fig. 6. Representative failure cases. Mode 1: rare pathology — attention correctly localised but the classifier under-fires due to class imbalance. Mode 2: overlapping pathologies — attention spreads across competing anatomical regions. Mode 3: dataset transfer — cross-domain label mismatches cause systematic under-prediction on specific OpenI labels.

5 Conclusion

We presented FAR-VLM, a faithfulness-aware vision-language model for chest X-ray diagnosis that integrates ViT-B/16 visual encoding, BioBERT textual encoding, and cross-modal attention fusion under a novel Faithfulness-Aware Regularisation loss. The FAR loss maximises the expected absolute difference between model confidence on original and patch-masked images, explicitly rewarding attention patterns that are causally important to predictions. Experiments on MIMIC-CXR demonstrate that FAR-VLM achieves $\text{AUC} = 0.9039$ and $\text{Accuracy} = 0.8804$, outperforming DenseNet-121 by 12.7% and 18.0% respectively, while attaining Deletion $\text{AUC} = 0.806$ — three times the faithfulness score of the vision-only baseline. Bootstrap 95% confidence intervals on the $k=50, \lambda=0.1$ configuration confirm $\text{AUC} = 0.814$ [0.809, 0.819], establishing statistical significance, and external validation on Indiana University OpenI yields $\text{AUC} = 0.759$, confirming cross-dataset generalisability. These results establish that faithfulness and classification performance are not competing objectives: the FAR loss improves both simultaneously by discouraging spurious attention patterns.

The present study has three limitations that motivate future work. First, the FAR-VLM F1-score (0.3690) reflects the inherent class imbalance in MIMIC-CXR; class-reweighted training or focal-loss modifications may improve this further. Second, CMFS relies on a manually curated anatomy keyword dictionary; a learned alignment model (e.g., a fine-tuned contrastive text-image encoder) could provide more robust cross-modal grounding. Third, external validation on OpenI (1,000 samples) is limited by the smaller dataset size and a different labelling protocol; a prospective validation on a held-out clinical dataset, together with a formal radiologist reader study, would further strengthen generalisability and clinical-utility claims.

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